Budapest University of Technology and Economics

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Bachelor Thesis

Zarankiewicz Numbers and Bipartite Ramsey Numbers

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Abstract

This thesis project enhances the computation of upper bounds for Zarankiewicz numbers through the development of an advanced Integer Linear Programming (ILP) model using the GLPK solver. The project begins by describing the foundational ideas and key bounds from previous studies. Building on these foundations, this thesis generalizes these bounds in a more comprehensive manner. Unlike previous works that focused on specific cases, such as small degrees in Collins et al. [1] idea and maximum degrees in Damásdi et al. [2] approach, this project employs more sharped constraints in recursion to achieve potentially better bounds. Furthermore, this project explores the relationship between Zarankiewicz numbers and bipartite Ramsey numbers, demonstrating how the Zarankiewicz numbers can help in establishing the upper bounds for the bipartite Ramsey numbers. The main contribution of this work is to improve the results of upper bounds for Zarankiewicz numbers by developing an advanced integer linear programming (ILP) model using the GLPK solver, which generalizes the bounds, making the process more flexible and efficient, surpassing previous methods like those of Irving [3] and Roman. This thesis contributes significantly to extremal graph theory by providing a robust computational tool and advancing the theoretical understanding of these bipartite graphs.

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Chapter 1

Introduction

1.1 Zarankiewicz numbers

1.1.1 Definition and Significance

A Zarankiewicz number $Z_{a,b}(x, y)$, represents the maximum number of edges in a complete bipartite graph $K_{x,y}$ with vertex classes of size x and y which contains no a vertices in the vertex class of size x and no b vertices in the vertex class of size y which span $K_{a,b}$ as subgraph. The order of a and b is crucial because it is not always the case that we get the same results. If $x = y$ and $a = b$, it is often called as diagonal or balanced, denoted as $Z_{a,a}(x, x)$. Let us begin with the basic methodology in order to understand how Zarankiewicz numbers are, actually, derived for very small parameters. The goal is to maximize a bipartite graph's edge count while preventing the creation of a $K_{a,b}$ subgraph. This requires us to be sure that no subset of a vertices from the first partite set A is connected to a subset of b vertices from the second partite set B in the given complete bipartite graph. A common approach is to construct bipartite graphs incrementally and verify whether adding an edge creates a forbidden $K_{a,b}$ subgraph. If it does, that edge is not added. If it doesn't, the edge is added, and the process continues until no more edges can be added without violating the $K_{a,b}$-free condition. Detailed methods and specific values for larger parameters will be explored in later chapters, providing a deeper understanding of how these numbers are computed and their implications in extremal graph theory.

The Zarankiewicz's problem is important in studying the extremal nature of bipartite graphs, which helps us in understanding their combinatorial structure better. Note that $Z_{a,b}(x, y) = Z_{b,a}(y, x)$ A more recent understanding of Chen, Horsley, and Mammoliti [4] suggests that if we consider a hypergraph H with vertex set of $V(H)$ and a multiset

$E(H)$ of edges such that each edge is a nonempty subset of $V(H)$, then $Z_{a,b}(x, y)$ can be defined as the maximum total degree $\sum_{e \in E(H)} |e|$ of hypergraph with x vertices and y edges, where each set of a vertices is a subset of at most $(b - 1)$ edges. In this way, we actually restrict each subset of a vertices to at most $(b - 1)$ edges by avoiding the existence of the complete bipartite subgraph. If b or more edges contains any subset of a vertices, it is possible to get a complete bipartite subgraph, which violates the condition of non-existence of $K_{a,b}$ in $K_{x,y}$.

1.1.2 Historical Context

The study of Zarankiewicz numbers arises from a fundamental problem in extremal graph theory, originally posed by Kazimierz Zarankiewicz in 1951. His initial question was focused on identifying the maximum number of edges in a bipartite graph $K_{x,x}$ avoiding $K_{a,a}$ as a subgraph. This problem was later generalized, and several upper and exact bounds for parameters satisfying specific conditions were shown using different methods. Sometimes they were asymptotically proved, see [5], [6], and [7] for a comprehensive overview of early results and extensions.

1.1.3 Recent Developments

Numerous developments have contributed to the understanding and evaluation of Zarankiewicz numbers. Notable contributions include the use of linear programming algorithms and computational methods to provide bounds for these numbers. Füredi's work [8] in 1996 introduced new asymptotic bounds for bipartite Turán numbers [7], which are similar to Zarankiewicz numbers. Valenzuela [9], in 2006, reported exact values for specified scenarios and described the relevant extremal bipartite graphs, and Nikiforov [10] used spectral radius approaches to improve prior results by Kovári, Sós, Turán, and Füredi. These results add to a better understanding of the upper bounds and help in the identification of near-optimal structures. Moreover, Damásdi, Héger, and Szőnyi [2] investigated the relationship between Zarankiewicz numbers and finite geometries, resulting in new constructions and bounds for specific parameters related to projective planes. For more detailed exploration of recent developments and applications, see the works of Collins et al. [1], Goddard, Henning, Oellermann [11], Conlon [12], and Chen, Horsley, and Mammoliti [4].

1.2 Bipartite Ramsey numbers

1.2.1 Definition and Significance

The bipartite Ramsey number, denoted as $R = R(a, b)$ is the smallest number of vertices R such that any bipartite graph $K_{R,R}$ whose edges are colored with two colors contains a monochromatic subgraph $K_{a,a}$ of the first color or $K_{b,b}$ of the second color. To understand how bipartite Ramsey numbers are truly formed for very small parameters, let's start with the basic approach. The goal is to determine the minimum number of vertices, R, in the corresponding complete bipartite graph $K_{R,R}$ such that any edge coloring of $K_{R,R}$ with two colors will necessarily contain a monochromatic $K_{a,b}$ in one of the colors. This requires us to ensure that any possible 2-coloring of the edges of $K_{R,R}$ will force a subset of a vertices from one vertex class A to be connected to a subset of b vertices from the other vertex class B in a single color. A common approach is to construct complete bipartite graphs gradually, starting with small values of R, and then examine all possible 2-colorings of the edges. For each 2-coloring, we check whether there exists a monochromatic $K_{a,b}$ subgraph in either color. If such a subgraph is found in every coloring, then R is considered enough for $R(a, b)$. If no monochromatic $K_{a,b}$ is found for a given R, we increase R and repeat the process until the condition is satisfied. The smallest R for which this holds true in all 2-colorings is the bipartite Ramsey number $R(a, b)$. When $a = b$ then we call this the balanced or diagonal, denoted by $R(a, a)$. Detailed methods and specific values for larger parameters will be explored in later chapters, providing a deeper understanding of how these numbers are computed and their implications in Ramsey theory. The importance of bipartite Ramsey numbers follows from their ability to generalize and extend the concepts from classical Ramsey theory to bipartite graphs. Over the time, many changes have been made in the definition of bipartite Ramsey numbers. Beineke and Schwenk [13] introduced definition for non-diagonal case as the minimal integer R that is specified in such a way that guarantees the existence of a monochromatic $K_{a,b}$ for any two coloring of the edges of $K_{R,R}$.

1.2.2 Historical Context

The study of bipartite Ramsey numbers has developed over time, with various exact values and bounds established via combinatorial methods. Actually, bipartite Ramsey numbers were derived from classical Ramsey numbers by bringing the notion of ensuring monochromatic subgraphs to the bipartite case. Moreover, Zarankiewicz numbers have been applied to establish upper bounds for bipartite Ramsey numbers. For a detailed overview of early results and the development of bipartite Ramsey theory, see the Irving's contribution [3].

1.2.3 Recent Developments

Recent progress in the study of bipartite Ramsey numbers has been motivated by both theoretical concepts and computational methods. There are two main research methods that are frequently used in the study of bipartite Ramsey numbers. The first approach focuses on establishing or improving upper and lower bounds for these numbers, which include hypergraph Ramsey numbers and Zarankiewicz numbers. This method is important for understanding the bounds for which parameters these numbers can exist. The second approach aims to determine exact values of these numbers, which is usually considered as impossible for large parameters because of the complexity involved. On the other hand, theoretical results can produce accurate bipartite Ramsey numbers for certain parameter values, providing valuable insights. Furthermore, computational techniques can be used to get accurate results for small parameters. Together, these approaches improve our understanding of bipartite Ramsey numbers and make a substantial contribution to its advancement. For example, Collins et al. [1] and Goddard, Henning, Oellermann [11] made significant improvements by offering new bounds and results on bipartite Ramsey numbers using some combinatorial methods.

Chapter 2

Zarankiewicz numbers

2.1 Bounding Zarankiewicz numbers

2.1.1 Upper bounds

Star Counting:

The basic idea that is widely used in most of the approaches in establishing the upper bound is star counting technique. The first, celebrated upper bound on $Z_{a,a}(x,x)$, Kővári, Sós and Turán [7] uses this idea. Moreover, Roman [6], Collins et al. [1], and Goddard, Henning, Oellermann [11] have used the same method, basically, it's about estimating the number of edges of a subgraph of $K_{x,y}$ not containing a $K_{a,b}$. The key idea is that we can calculate the number of $K_{1,b}$ subgraphs from the degrees of the vertices.

To see how this approach actually works, we first consider a bipartite graph G with partite sets A and B of cardinality x and y, respectively, and suppose that G does not contain a $K_{a,b}$. Let $v_1, v_2, \dots, v_x$ be the vertices in set A, and $u_1, u_2, \dots, u_y$ be the vertices in set B.

Given this, let $G = (A \cup B, E)$ be a bipartite graph with vertex classes in the given order. We define $N(G, K_{a,b})$ as the number of subgraphs of G that are isomorphic to $K_{a,b}$, where a vertices are in set A and b vertices are in set B. Once G is fixed, we can use $N_{K_{a,b}}$ for short.

Let v_i be a vertex from set A with degree $d(v_i)$, where $i = 1, \dots, x$. Let b be the number of vertices from set B that are incident to vertex v_i. Then, from set A,

$$N(G, K_{1,b}) = \sum_{i=1}^{x} \binom{d(v_i)}{b},$$

where 1 in $K_{1,b}$ represents the vertex v_i. This is true because any set of b vertices in B can be in at most $(a-1)$ distinct $K_{1,b}$ subgraphs; otherwise, we would obtain a $K_{a,b}$ subgraph.

Similarly, from set B we have $K_{1,b} \leq (a-1)\binom{y}{b}$, so we get

$$\sum_{i=1}^{x} \binom{d(v_i)}{b} \leq (a-1)\binom{y}{b}. \tag{2.1}$$

Thus, the number of edges $|E|$ in the bipartite graph $K_{x,y}$ is given by: $|E| = \sum_{i=1}^{x} d(v_i)$, where $d(v_i)$ is the degree of vertex v_i in set A. This sum can be large if the degrees $d(v_i)$ are sufficiently large. However, if (2.1) leads to a contradiction, it implies that $|E|$ must be smaller. In other words, if there is no subgraph $K_{a,b}$ in G, the sum $\sum_{i=1}^{x} \binom{d(v_i)}{b}$ must be bounded in such a way that the total number of edges $|E|$ does not exceed certain limits.

Proposition 1. ([1]) Let d, k, and t be the fixed positive integers then among the k part sums $v_1 + v_2 + ... + v_k = d$ with $v_i \geq 0$, the sum

$$\sum_{i=1}^{k} \binom{v_i}{t} \tag{2.2}$$

is minimum when $|v_i - v_j| \leq 1$ for all $1 \leq i < j \leq k$.

Proof. Let $v_1 + v_2 + ... + v_k = d$, and suppose $|v_i - v_j| \leq 1$ for all $1 \leq i < j \leq k$ is not true, then consider without loss of generality that $v_1 \geq v_2 \geq ... \geq v_k$ and $v_i - v_j > 1$. We define $u_1 = v_1 - 1, u_2 = v_2, ..., u_{k-1} = v_{k-1}, u_k := v_k + 1$ be a new partition of d into k parts. Then, we have:

$$\sum_{i=1}^{k} \binom{v_i}{t} - \sum_{i=1}^{k} \binom{u_i}{t} = \binom{v_1}{t} - \binom{v_1 - 1}{t} + \binom{v_k}{t} - \binom{v_k + 1}{t}.$$

We will show that this difference is non-negative. To see this we use the property of binomial coefficients:

$$\binom{v}{t} = \binom{v-1}{t} + \binom{v-1}{t-1}.$$

Therefore,

$$\binom{v_1}{t} - \binom{v_1 - 1}{t} = \binom{v_1 - 1}{t-1}$$

and

$$\binom{v_k + 1}{t} - \binom{v_k}{t} = \binom{v_k}{t-1}.$$

for $v_1 > v_{k+1}$, so $v_1 - 1 > v_k$, we know that the sum of these modifications will be non-negative:

$$\binom{v_1}{t} - \binom{v_1 - 1}{t} + \binom{v_k}{t} - \binom{v_k + 1}{t} = \binom{v_1 - 1}{t-1} - \binom{v_k}{t-1} \geq 0,$$

as $v_1 - 1 > v_k$. Morever, if $t - 1 > 0$, so $t \geq 2$, then this is strictly positive. Thus, the change initiates to a new partition of d where the sum of the binomial coefficients does not increase. Repeating this modification, we arrive in the situation we stated ($|v_i - v_j| \leq 1$, for all i, j), and the sum in (2.2) never increased, so we are done. resulting that the original (2.2) sum is minimized.

□

Example 1. $Z_{2,2}(7,6) \leq 18$.

To show that this is really satisfied we assume that $Z_{2,2}(7,6) \geq 19$. This means that there exist a bipartite graph $G(A,B,E)$ with $|A| = 7$, $|B| = 6$, $|E| = 19$ and no $K_{2,2}$ as subgraph. Then,

$$|E| = \sum_{i=1}^{7} d(v_i) = 19,$$

and from (2.1)

$$\sum_{i=1}^{7} \binom{d(v_i)}{2} \leq \binom{6}{2} = 15.$$

We determine the smallest possible value of $\binom{d(v_i)}{2}$ given that $\sum_{i=1}^{7} d(v_i) = 19$. Since $d(v_1) + d(v_2) + \ldots + d(v_k) = 19$, we get possible values as: $d(v_i) = \left\lfloor \frac{19}{7} \right\rfloor = 2$ or $d(v_i) = \left\lceil \frac{19}{7} \right\rceil = 3$. Then, we can write 19 as: $19 = 2 \cdot 7 + 5$, thus 5 of the $d(v_i)$'s $= 3$ and 2 of the $d(v_i)$'s $= 2$ must hold, if $|v_i - v_j| \leq 1$ for all i, j. Therefore, $\sum_{i=1}^{7} \binom{d(v_i)}{2}$ is the minimal for $5 \cdot \binom{3}{2} + 2 \cdot \binom{2}{2} = 15 + 2 = 17$. However, by (2.1), this should be at most 15, which is contradiction. Hence, $Z_{2,2}(7,6) \leq 18$.

Next, we give Roman's upper bound, which uses the same idea, but introduces a parameter p which makes the formula easy to use.

Theorem 1. (Roman [6]) If $Z_{a,b}(x,y)$ is the Zarankiewicz number, then

$$Z_{a,b}(x,y) \leq \frac{b-1}{\binom{p}{a-1}} \cdot \binom{x}{a} + \frac{(p+1)(a-1)}{a} \cdot y, \tag{2.3}$$

for all integers $p \geq a - 1$.

Proof. Let G be a bipartite graph of two partite sets A and B with $Z = Z_{a,b}(x, y)$ edges such that it doesn't contain $K_{a,b}$ as a subgraph. To prove (2.3), let $v_1, v_2, \ldots, v_x$ be the vertices in set A, and $u_1, u_2, \ldots, u_y$ be the vertices in set B. Let $\sum_{i=1}^{y} d_{u_i} = Z$ be the number of edges incident to the i^{th} vertex from the partite set B in the graph G. Then, $Z_{a,b}(x, y)$ equals the sum of all e_i. If all vertices in the set B have less than $a-1$ incident edges, then by adding edges to that vertex arbitrarily until it has at least $a-1$ incident edges such that it doesn't become a complete subgraph $K_{a,b}$. Thus, we can assume that $e_i \geq a-1$ for all $i = 1, 2, \ldots, y$.

Let $\sum_{i=1}^{y} d(u_i) = Z$ be the number of edges, where $d(u_i)$ represents the degree of u_i vertex from set A incident to the vertices of set B in the graph G. If a vertex in set B have fewer than $a-1$ incident edges, then by adding edges to those vertices arbitrarily until it has exactly $a-1$ incident edges, we still don't get a complete subgraph $K_{a,b}$. Thus, we can assume that $d(u_i) \geq a-1$ for all $i = 1, 2, \ldots, y$.

By star counting (that is (2.1)), we get

$$\sum_{i=1}^{y} \binom{d_{ui}}{a} \leq (b-1)\binom{x}{a}. \tag{2.4}$$

Then, for $0 < \alpha \in \mathbb{R}$, (2.4) gives,

$$\sum_{i=1}^{y} \alpha e_i(e_i-1)...(e_i-a+1) \leq \alpha a!(b-1)\binom{x}{a}$$

here, since $Z = Z_{a,b}(x, y)$ equals to the sum of all d_{ui}, i.e. $\sum_{i=1}^{y} e_i$, so we get

$$\sum_{i=1}^{y}[\alpha e_i(e_i-1)...(e_i-a+1)-e_i] + Z \leq \alpha a!(b-1)\binom{x}{a}$$

and for any $\beta \in \mathbb{R}$

$$\sum_{i=1}^{y}[\alpha e_i(e_i-1)...(e_i-a+1)-e_i+\beta] + Z \leq \alpha a!(b-1)\binom{x}{a} + \beta y \tag{2.5}$$

Now, we will begin by considering the polynomial

$$f(\xi) = \alpha\xi(\xi-1)...(\xi-a+1) - \xi + \beta.$$

If we can show that, for suitable selections of α and β, $f(\xi)$ is positive for all integer values of ξ, then we will get from (2.5) that

$$Z \leq \alpha a!(b-1)\binom{x}{a} + \beta y \tag{2.6}$$

To do this, for any integer $p \geq a-1$, we select values for α and β such that $f(p) = 0$ and $f(p+1) = 0$. By $f(p) = f(p+1) = 0$, we get

$$\alpha p(p-1)...(p-a+1) - p + \beta = 0,$$

$$\alpha(p+1)p(p-1)...(p-a+2) - (p+1) + \beta = 0,$$

therefore,

$$\alpha = \frac{1}{ap(p-1)...(p-a+2)} \; ; \; \beta = \frac{(a-1)(p+1)}{a}.$$

By substituting α and β into the original equations, it can be verified that both equations are satisfied, confirming the correctness of these values.

Finally, we claim that $f(\xi) \geq 0 \; \forall \xi \in \mathbb{Z}$ such that $\xi \geq a-1$,. Now, We verify that $f(\xi)$ is convex for $\xi \geq a-1$. To see this, let us calculate its derivative:

$$f'(\xi) = \alpha \sum_{j=1}^{a} 1^a \xi(\xi-1)... \overbrace{(\xi-j+1)}^{j^{th}\text{-term}} ...(\xi-a+1) - 1,$$

where we delete the j^{th}-term in the sum. Here, as $\alpha > 0$, we can observe that $f'(\xi)$ is an increasing function for $\xi \geq a-1$. Therefore, $f(\xi)$ is a convex function for $\xi \geq a-1$ and its only two roots are consecutive integers. Based on these observations, we can say that our claim is established. Hence, we can say that for $\alpha = \frac{1}{ap(p-1)...(p-a+2)}$ and $\beta = \frac{(a-1)(p+1)}{a}$ we have

$$Z \leq \alpha a!(b-1)\binom{x}{a} + \beta y$$

i.e.,

$$Z_{a,b}(x,y) \leq \frac{b-1}{\binom{p}{a-1}} \cdot \binom{x}{a} + \frac{(p+1)(a-1)}{a} \cdot y$$

□

Example 2. Let's take the same parameters as before: $Z_{2,2}(7,6) \leq 18$.

We apply Roman's bound (2.3). Note that Roman's bound can give different results for different arrangements of parameters, so we check for which order it gives the best result, and we choose that as the upper bound. Since this is the balanced case ($a = b$), therefore, we only have two possible variations and that is when we will only interchange the x and y.

Case 1: $Z_{a,b}(x, y)$ $x = 7, y = 6, a = b = 2$, and $p = 2$, where p is the floor of the expected average degree, but can be anything as well. Now by using Roman's bound (2.3) for given parameters we get:

$$Z_{2,2}(7, 6) \leq \frac{2-1}{2} \cdot 21 + \frac{(2+1)(2-1)}{2} \cdot 6 = 10.5 + 9 = 19.5$$

Case 2: $Z_{a,b}(y, x)$ Here we interchange the order of x and y, then we get:

$$Z_{2,2}(6, 7) \leq \frac{2-1}{2} \cdot 15 + \frac{(2+1)(2-1)}{2} \cdot 7 = 7.5 + 10.5 = 18$$

Consequently, we can say that 18 is an upper bound for $Z_{2,2}(7, 6)$.

Equality of Roman's bound:

Finding equality for Zarankiewicz numbers is very challenging task, however, by using design theory, one can determine exact bounds of Zarankiewicz numbers. When certain combinatorial designs exist for some parameters, then by using their corresponding incidence graphs we can find the maximum number of edges without having $K_{a,b}$ as subgraph. We will see now for which cases equality on Roman's bound hold. In addition, further information on the design terminology and how it relates to Zarankiewicz numbers can be found in [2] and [6]. Notably, the study provides precise numbers and examples that show how design theory might be applied to these exact bounds.

By examining the proof of the theorem (2.3), equality holds in this bound for a particular value of p whenever there is a complete bipartite graph $K_{x,y}$ with vertex classes of size x and y such that it avoids containing $K_{a,b}$ as subgraph, and the sum of binomial coefficients $\binom{d_i}{a}$ over all y vertices in one partite set equals $(b-1)\binom{x}{a}$, i.e. $\sum_{i=1}^{y} \binom{d_i}{a} = (b-1)\binom{x}{a}$; where each vertex in this partite set must have degree p or $p + 1$.

A t-design, denoted as t-(v, k, λ), is a class of k-element subsets of X, called *blocks*, such that every t-element subset T appears in exactly λ blocks. The parameters of the design are v (the number of elements of X), b (the number of blocks), k, λ, and t.

Equality holds in the bound for a particular p whenever the t-design t-(v, k, λ) exists and y equals the number of blocks in the design, b. Specifically, we are interested in the configurations where $t = a$, $v = x$, $k = p$ or $p + 1$, and $\lambda = b - 1$.

Wilson [14] has shown that given positive integers p and $b-1$, a 2-(v,k,λ) design exists for all sufficiently large x satisfying:

$$(b-1)(x-1) \equiv 0 \pmod{p-1},$$

$$(b-1)x(x-1) \equiv 0 \pmod{p(p-1)}.$$

This means that for any two positive integers p and $b-1$, if x is sufficiently large and satisfies the above conditions, then

$$Z_{a,b}(x,y) = \frac{b-1}{\binom{p}{a-1}} \cdot \binom{x}{a} + \frac{(p+1)(a-1)}{a} \cdot y,$$

for $y = \frac{(b-1)}{\binom{p}{a-1}} \cdot \binom{x}{a}$. Thus, equality holds for infinitely many values of x and y.

Additionally, Hanani [15] has shown that, except for a 2-$(15,5,2)$ design which does not exist, if $3 \leq p < 5$, a necessary and sufficient condition for the existence of a 2-$(x,p,p-1)$ design is that the above modular conditions are satisfied. Hence, equality holds for $a=2$, b arbitrary, and $3 \leq p < 5$ whenever x satisfies the modular conditions and $y = \frac{(b-1)}{\binom{p}{a-1}} \cdot \binom{x}{a}$.

Next we give Irving's improvement of the star counting method. When counting stars, we use the information about the number of $K_{a,b}$ subgraphs (which is zero) to derive our upper bound on the number of edges (that is $K_{1,1}$ subgraphs). Irving generalizes this: given a lower bound on the number of $K_{s,t}$ subgraphs, he derives a lower bound on the number of $K_{s,w}$ subgraphs, for more general parameters s,t and w. The point is that if the number of edges ($K_{1,1}$ subgraphs) is large, using subsequent estimates, we end up with a positive lower bound on the number of $K_{a,b}$ subraphs, a contradiction.

Lemma 1. (Irving [3]) Suppose that, in a subgraph of $K_{x,y}$, the number of copies of $K_{s,t}$, denoted by $N_{K_{s,t}}$, is at least e. Then

(i) the number of copies of $K_{s,w}$ $(t<w)$ in the subgraph is at least

$$N_{K_{s,w}} \geq \min_{d_i} \sum_{i=1}^{\binom{x}{s}} \binom{d_i}{w}$$

where the d_i $\left(1 \leq i \leq \binom{x}{s}\right)$ are non-negative integers subject to

$$\sum_{i=1}^{\binom{x}{s}} \binom{d_i}{t} \geq e.$$

(ii) The number of copies of $K_{w,t}$ $(s < w)$ in the subgraph is at least

$$N_{K_{w,t}} \geq \min_{d_i} \sum_{i=1}^{\binom{y}{t}} \binom{d_i}{w}$$

where the d_i $\left(1 \leq i \leq \binom{y}{t}\right)$ are non-negative integers subject to

$$\sum_{i=1}^{\binom{y}{t}} \binom{d_i}{s} \geq e.$$

Proof. (i) Let A and B denote respectively the x-vertex set and the y-vertex set. Let d_i $\left(1 \leq i \leq \binom{x}{s}\right)$ denote the number of vertices of B joined by an edge to each of the vertices in the i-th a-subset of A.

To count the number of $K_{s,t}$ subgraphs, we note that forming a $K_{s,t}$ involves selecting t vertices from the d_i vertices in B connected to an s-subset of A, and the number of ways to select t vertices from d_i vertices is given by $\binom{d_i}{t}$. Summing the number of ways to form $K_{s,t}$ subgraphs for each of the $\binom{x}{s}$ possible s-subsets of A gives us the total number of $K_{s,t}$ subgraphs in the bipartite graph, as $N_{K_{s,t}} \geq e$, we obtain the constraint

$$\sum_{i=1}^{\binom{x}{s}} \binom{d_i}{t} \geq e.$$

Now, consider the number of $K_{s,w}$ subgraphs. Forming a $K_{s,w}$ involves selecting w vertices from the d_i vertices in B connected to an s-subset of A, and the number of ways to select w vertices from d_i vertices is given by $\binom{d_i}{w}$. Therefore, the total number of $K_{s,w}$ subgraphs is:

$$\sum_{i=1}^{\binom{x}{s}} \binom{d_i}{w}.$$

Since we are looking for the minimum number of such $K_{s,w}$ subgraphs, we consider the minimum value of this sum under the constraint $\sum_{i=1}^{\binom{x}{s}} \binom{d_i}{t} \geq e$. Thus, the number of copies of $K_{s,w}$ is at least:

$$N_{K_{s,w}} \geq \min_{d_i} \sum_{i=1}^{\binom{x}{s}} \binom{d_i}{w}.$$

(ii) Similarly, let A and B denote respectively the x-vertex set and the y-vertex set. Let d_i $\left(1 \leq i \leq \binom{y}{t}\right)$ denote the number of vertices of A joined by an edge to each of the vertices in the i-th t-subset of B.

To count the number of $K_{s,t}$ subgraphs, we note that forming a $K_{s,t}$ involves selecting s vertices from the d_i vertices in A connected to a t-subset of B, and the number of ways to select s vertices from d_i vertices is given by $\binom{d_i}{s}$. Summing the number of ways to form $K_{s,t}$ subgraphs for each of the $\binom{y}{t}$ possible t-subsets of B gives us the total number of $K_{s,t}$ subgraphs in the bipartite graph, implying that the number of copies $N_{K_{s,t}}$ is at least e:

$$\sum_{i=1}^{\binom{y}{t}} \binom{d_i}{s} \geq e.$$

Now, consider the number of $K_{w,t}$ subgraphs. Forming a $K_{w,t}$ involves selecting w vertices from the d_i vertices in A connected to a t-subset of B, and the number of ways to select w vertices from d_i vertices is given by $\binom{d_i}{w}$. Therefore, the total number of $K_{w,t}$ subgraphs is:

$$\sum_{i=1}^{\binom{y}{t}} \binom{d_i}{w}.$$

Since we are looking for the minimum number of such $K_{w,t}$ subgraphs, we consider the minimum value of this sum under the constraint $\sum_{i=1}^{\binom{y}{t}} \binom{d_i}{s} \geq e$. Thus, the number of copies of $K_{w,t}$ is at least:

$$N_{K_{w,t}} \geq \min_{d_i} \sum_{i=1}^{\binom{y}{t}} \binom{d_i}{w}.$$

□

Consequently, starting with the given number of edges e, by using lemma repeatedly we can determine the minimum number of a particular kind of subgraph in the $K_{x,x}$. As we consider larger and larger bicliques, we eventually can prove the existence of the spanning subgraph G in $K_{x,x}$.

In extremal graph theory, recursive method is widely used to determine the upper bound of Zarankiewicz numbers. For instance, Collins et al. [1] approach uses the recursive technique that determines the maximum number of edges while avoiding certain substructures. The following straightforward result from [2] in this area plays an important role in understanding the recursive method. It is useful for generating tighter bounds.

Proposition 2. (Dámasdi, Héger, and Szőnyi [2])

Let $U_{a,b}(x,y,s,t) = Z_{a-s,t}(x-s,y-t) + Z_{a,b}(x-s,y-t) + (s-1)y$. Then

$$Z_{a,b}(x,y) \leq \min_s \max_t \min \{Z_{s,t+1}(x,y), U_{a,b}(x,y,s,t) : 1 \leq s < a, b-1 \leq t \leq y\}.$$

Proof. To prove this, consider a maximal $K_{a,b}$-free bipartite graph $G = (A,B;E)$ with $x+y$ vertices. First, select s such that $1 \leq s < a$ and let t be the largest integer for which the bipartite subgraph $K_{s,t}$ is contained in G. Since G is $K_{s,t+1}$-free by the maximality of t, therefore, the number of edges $|E|$ in G is at most $Z_{s,t+1}(x,y)$. Note that the ordering of the vertex classes does matter in this context.

Next, choose subsets $S \subset A$ and $T \subset B$ such that the induced subgraph on S and T is $K_{s,t}$. Define the sets $M = A \setminus S$ and $N = B \setminus T$. The subgraph induced on M and T, denoted $G[M,T]$, must be $K_{a-s,b}$-free becasue if M has $a-s$ vertices and T has $b-1$ vertices, maximality, then $G[M,T]$ will be $K_{a-s,b}$-freew. Similarly, the subgraph induced on M and N, denoted $G[M,N]$, must be $K_{a,b}$-free.

Given that no $K_{s,t+1}$ can be found in G, each vertex in N can have at most $s-1$ neighbors in S. By summing the maximum number of edges in each part of the graph, we obtain the inequality $|E| \leq st + Z_{a-s,b}(x-s,t) + Z_{a,b}(x-s,y-t) + (s-1)(y-t)$. This sum is exactly the definition of $U_{a,b}(x,y,s,t)$.

As G is a maximal $K_{a,b}$-free graph, it must contain a $K_{s,b-1}$ for all $s < a$. Therefore, t must be at least $b-1$. This completes the proof, showing that the inequality $Z_{a,b}(x,y) \leq \min_s \max_t \min\{Z_{s,t+1}(x,y), U_{a,b}(x,y,s,t)\}$ holds for the given parameters.

□

2.1.2 Some small values

In this section, we consider small examples with known exact values of Zarankiewicz numbers. Understanding these small cases of Zarakiewicz numbers provides the foundation underlying this whole notion. To demonstrate Zarankiewicz numbers for small parameters, we will begin by looking at the exact value from Collins et al. [1] for $Z_{2,2}(5,5) = 12$.

In figure (2.1), we show a bipartite graph with two sets of vertices, each of cardinality five i.e. $X = 5$ and $Y = 5$. The edges are drawn to maximize the number of edges while ensuring no $K_{2,2}$ subgraph exists within the bipartite graph. We drew 12 edges between vertices of X and Y to avoid creating a complete bipartite subgraph $K_{2,2}$. Adding more edges between would create a $K_{2,2}$ subgraph, violating the definition. To see how we got this 12 edges as maximum, suppose there is a graph on 5 by 5 vertices, 13 edges, no

$K_{2,2}$. Then, $a_1, \ldots, a_5$ are the vertices in A. $d(a_1) = 5$ cannot hold. Assume $d(a_1) \geq \ldots \geq d(a_5)$. Otherwise, every $d(a_i) \leq 1$ $(i \neq 1)$, so $\leq 5 + 4 = 9$ edges. If $d(a_1) = 4$, then $a_2 + a_3 + \ldots + a_5 = 9$, hence $d(a_2) \geq 3$, but then a_1 and a_2 have at least 2 common neighbors $K_{2,2}$, contradiction. Thus $d(a_1) = 3$. As $d(a_2) + \ldots + d(a_5) = 10$, $d(a_2) = d(a_3) = 3$. Again we find a $K_{2,2}$.

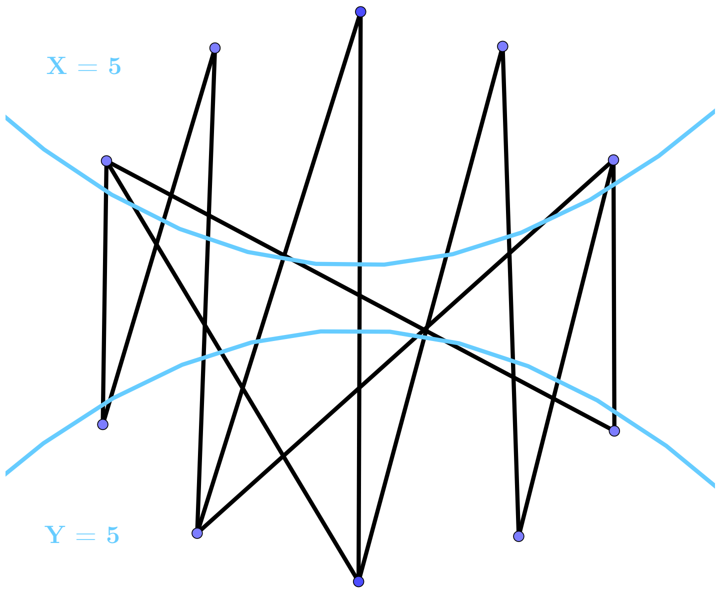

Figure 2.1: Bipartite graph of $Z_{2,2}(5, 5) = 12$

The exact values of Zarankiewicz numbers for small parameters are important in determining the Zarankiewicz number for the larger parameters. For a comprehensive overview of smaller and exact values of Zarankiewicz numbers, including detailed methodologies and results, see the works of Collins et al. [1], Goddard, Henning, Oellermann [11], Chen, Horsley, and Mammoliti [4], and Irving [3].

Chapter 3

Bipartite Ramsey numbers

3.1 Bounding bipartite Ramsey numbers

One of the primary goals of this chapter is to identify the key boundaries that limit the presence of monochromatic subgraphs within complete bipartite graphs. The approach used to obtain these constraints is discussed in this section, along with their importance for understanding the small details associated with bipartite Ramsey numbers.

Hadamard matrix:

Hadamard matrices play a crucial role in understanding the bounds for bipartite Ramsey numbers. The structured nature of a Hadamard matrix establishes the construction of precise combinatorial analysis by proving exact values, lower bounds, and upper bounds. However, it is important to note that the existence of Hadamard Matrix of certain orders is still an open question for some cases.

3.1.1 Lower bounds

Beineke and Schwenk [13] established results of lower bounds for bipartite Ramsey numbers based on Hadamard matrices. Few results for $R(a, b)$ are given in [13]:

Theorem 2. R(2,b) $\geq 4b - 3$,
if there is Hadamard Matrix of order $2(b-1)$, or b is odd.

Proof. To prove this, let's assume there exists a Hadamard matrix H of order $2(b-1)$. From this matrix, we construct a larger matrix M by arranging H and $-H$ in a block pattern as follows:

$$M = \begin{pmatrix} H & -H \\ -H & H \end{pmatrix}.$$

This matrix M is of order $4(b-1)$. Let A be the set of vertices corresponding to the rows of M and B be the set of vertices corresponding to the columns of M. We define two bipartite graphs, G and G', as follows. In the graph G, an edge exists between a vertex a_i in A and a vertex b_j in B if the corresponding entry $M_{ij} = +1$. In the graph G', an edge exists between a vertex a_i in A and a vertex b_j in B if the corresponding entry $M_{ij} = -1$. Clearly, $G \cup G' = K_{4b-4,4b-4}$

The key property of Hadamard matrices is that for any two rows (or columns) of H, the dot product of these rows (or columns) is zero. This implies that in the matrix M, any two rows (or columns) will have exactly $2(b-1)$ common entries, half of which are $+1$ and the other half are -1. Consequently, for any pair of vertices in A, there will be exactly $b-1$ common neighbors in B corresponding to $+1$ entries in M, and exactly $b-1$ common neighbors in B corresponding to -1 entries in M.

This construction ensures that neither G nor G' can contain a complete bipartite graph $K_{2,b}$. If G or G' did contain $K_{2,b}$, there would be two rows (or columns) in M with more than $b-1$ common $+1$ or -1 entries, which is impossible given the properties of the Hadamard matrix.

Therefore, the construction of the bipartite graphs G and G' using the Hadamard matrix H ensures that $R(2,b) \geq 4b-3$.

□

Theorem 3. R(2,b) $\geq 4b-4$,
if there is Hadamard Matrix of order $4(b-1)$.

Proof. To prove that $R(2,b) \geq 4b-4$ given the existence of a Hadamard matrix of order $4(b-1)$, we proceed similarly to the proof of (2).

Assume there exists a Hadamard matrix H of order $4(b-1)$. Construct the matrix M by arranging H and $-H$ in a block pattern as we did previously. Here, this matrix M is of order $8(b-1)$. We define the vertex sets A and B corresponding to the rows and columns of M, respectively, and construct two bipartite graphs, G and G' as:

- In G, an edge exists between $a_i \in A$ and $b_j \in B$ if $M_{ij} = +1$.
- In G', an edge exists between $a_i \in A$ and $b_j \in B$ if $M_{ij} = -1$.

Now, using the orthogonality property of Hadamard matrices that is for any two rows (or columns) of H, the dot product is zero, meaning the number of positions where the entries are both $+1$ is equal to the number of positions where the entries are both -1. In M, any two rows (or columns) will have exactly $2(b-1)$ positions where the entries are both $+1$ and $2(b-1)$ positions where the entries are both -1.

Consequently, for any pair of vertices in A, there will be exactly $2(b-1)$ common neighbors in B for both $+1$ and -1 entries in M. This construction ensures that neither G nor G' can contain a complete bipartite graph $K_{2,b}$. If G or G' did contain $K_{2,b}$, it would mean that there are two vertices in A that are both connected to b vertices in B, implying more than $2(b-1)$ common neighbors, which contradicts the Hadamard matrix property. Therefore, the construction using the Hadamard matrix H ensures $R(2,b) \geq 4b-4$.

□

Theorem 4. $R(3,b) \geq 8b-7$,

if there is Hadamard Matrix of order $4(b-1)$.

Determining $R(3,b)$ is a much more difficult problem. The existence of a Hadamard matrix of order $4(b-1)$ can be shown to imply that $R(3,b) \geq 8b-7$.

In addition to the above results, Hattingh and Henning [16] established bounds using probabilistic method for balanced bipartite Ramsey number.

Theorem 5. $R(x,x) > \frac{\sqrt{2}}{e} \cdot x \cdot 2^{\frac{x}{2}}$,

for all sufficiently large x.

3.1.2 Upper bounds

For $x \geq 21$, Irving [3] proved that $R(x,x) < 2^{x-1}(x-1)$, however, Goddard, Henning, Oellermann [11] showed for unbalance case that this method is helpless when y grows larger than x. Hattingh and Henning [16] introduced upper bound:

Theorem 6. $R(a,b) \leq \binom{a+b}{a} - 1$

The bound $R(a,b) \leq \binom{a+b}{a} - 1$ provides a useful upper bound for bipartite Ramsey numbers. However, this is not the only approach to determining upper bounds. In a later chapter, specifically in the section on the relationship between Zarankiewicz numbers and bipartite Ramsey numbers (4), we will see another significant upper bound based on Zarankiewicz numbers.

3.1.3 Exact values

Exact values are rigorously determined for some bipartite Ramsey numbers $R(a, b)$ by Beineke and Schwenk [13]. They showed some results, especially, in case of $R(1, b)$, $R(2, b)$, and $R(3, b)$. Later Irving [3] contributed by improving bound for $R(3, b)$ that strengthened previously known result in [13]. Beineke and Schwenk showed that:

Theorem 7. $R(1, b) = 2b - 1$

Theorem 8. $R(2, b) = 4b - 3$,

if there is Hadamard matrix of order $2(b - 1)$, $b = 2$, or b is odd.

Theorem 9. $R(3, b) = 8b - 7$,

if there is Hadamard matrix of order $4(b - 1)$. This follows from [3] and [13].

Example 3. If there exist Hadamard matrix of order $12 \cdot 12$, .i.e. when b $= 4$, then from (9), we get:

$$R(3, 4) = 25$$

We can see the existence of such Hadamard matrix as:

$$M = \begin{pmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
-1 & 1 & -1 & 1 & -1 & -1 & -1 & 1 & 1 & 1 & -1 & 1 \\
-1 & 1 & 1 & -1 & 1 & -1 & -1 & -1 & 1 & 1 & 1 & -1 \\
-1 & -1 & 1 & 1 & -1 & 1 & -1 & -1 & -1 & 1 & 1 & 1 \\
-1 & 1 & -1 & 1 & 1 & -1 & 1 & -1 & -1 & -1 & 1 & 1 \\
-1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 & -1 & -1 & -1 & 1 \\
-1 & 1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 & -1 & -1 & -1 \\
-1 & -1 & 1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 & -1 & -1 \\
-1 & -1 & -1 & 1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 & -1 \\
-1 & -1 & -1 & -1 & 1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 \\
-1 & 1 & -1 & -1 & -1 & 1 & 1 & 1 & -1 & 1 & 1 & -1 \\
-1 & -1 & 1 & -1 & -1 & -1 & 1 & 1 & 1 & -1 & 1 & 1
\end{pmatrix}$$

The matrix shown above satisfies the important conditions for Hadamard matrix. First of all, it satisfy the orthogonality property, indicating that any two different rows (or columns) are orthogonal if their dot product is zero. Second, uniformity across all rows and columns is ensured by the matrix's constant row sums and constant column sums. Thirdly, all of the matrix's entries are $+1$ or -1, which satisfies with Hadamard matrices' required entries. These properties confirm that the matrix M is indeed a valid Hadamard matrix of order 12.

3.1.4 Some small values

In this section, we consider a example of exact value of bipartite Ramsey number with small parameters. Understanding these small value will serve as a foundation for larger parameters. Finding the smallest number R such that any two-coloring of the edges of the complete bipartite graph $K_{R,R}$ results in a monochromatic $K_{a,b}$. Here, we take example of $a = b = 2$. We know from [11] that $R(2,2) = 5$, but we first understand why it's not equal to 4.

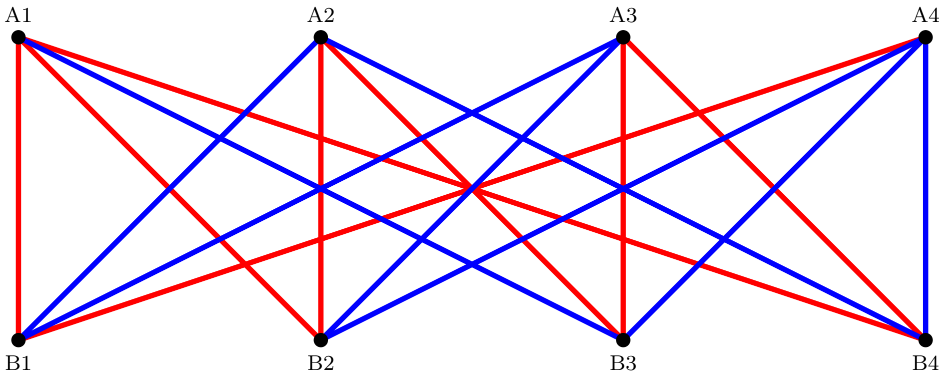

Figure 3.1: Red-blue coloring of $K_{4,4}$

Let the bipartite complete graph $K_{4,4}$ with red-blue coloring as shown in figure (3.1) with two sets of vertices, A and B, each having a cardinality of four. Set A's vertices are $A1$, $A2$, $A3$, and $A4$, while Set B's vertices are $B1$, $B2$, $B3$, and $B4$. Every vertex in set A is connected to every vertex in set B in this bipartite graph example. We construct the graph in such a way that no $K_{2,2}$ of any color appears. This indicates that no two vertices in set A and no two vertices in set B have four edges that are all the same color.

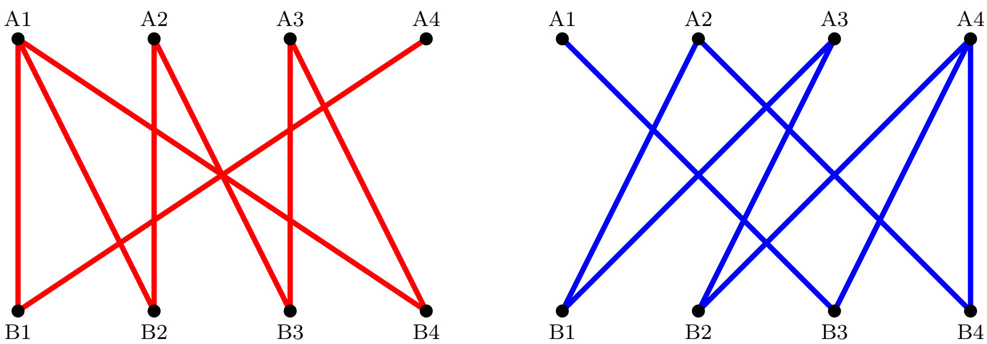

Figure 3.2: Red-blue subgraphs of $K_{4,4}$

We can observe that the edges in figure (3.1) are colored in a way that prevents the existence of monochromatic $K_{2,2}$. Now, to see how edge coloring in figure (3.1) is arranged to avoid the $K_{2,2}$ of any color, we consider these bipartite subgraphs (3.2). These subfigures actually help us to see which edges are colored blue and which are red in the previous bipartite graph (3.1).Thus, we can say $R(2,2) > 4$.

Corollary 1. $R(2,2)=5$

Proof. Let the complete bipartite graph $K_{5,5}$ as in (3.3) with two partite sets A and B, each consisting of five vertices. We color the edges of this graph with red and blue coloring. Let's assume that there is no biclique of $K_{2,2}$ of either color in the graph. We pick an arbitrary vertex let's say A_3 from set A, then we notice here that there are five edges going out from this vertex. Now, by pigeonhole principle, we can say that at least 3 of those incident edges to vertex A_3 are of the same color (red). Then, without loss of generality, we consider that the edges incident to vertex A_3 are connected to vertices B_1, B_3, B_4 from set B with red edges. Here, we observe that any two of these vertices, which share a common neighbor in vertex class A connected by edges of the same (red) color, will form a monochromatic red $K_{2,2}$, contradicting our initial assumption that no such subgraph exists. Therefore, every vertex A_i in A ($i \neq 3$ has at most 1 red edge, thus at least two blue edges going to B_1, B_3, B_4. As there is no blue $K_{2,2}$, no two vertices from A can have the same two neighbors on blue edges, but there are 4 vertices, A_1, A_2, A_4, and A_5, but only three ways to choose to vertices from B_1, B_3, B_4, so these must coincide for two A_i, A_j, hence a blue $K_{2,2}$ is formed. .Similarly, if we assume for blue edge we will get the contradiction. Therefore, any coloring of the edges in $K(5,5)$ leads to a monochromatic $K(2,2)$ subgraph, our assumption must be false, meaning any bipartite graph $K(5,5)$ must contain a monochromatic $K(2,2)$ subgraph. (see figure (3.3)).

□

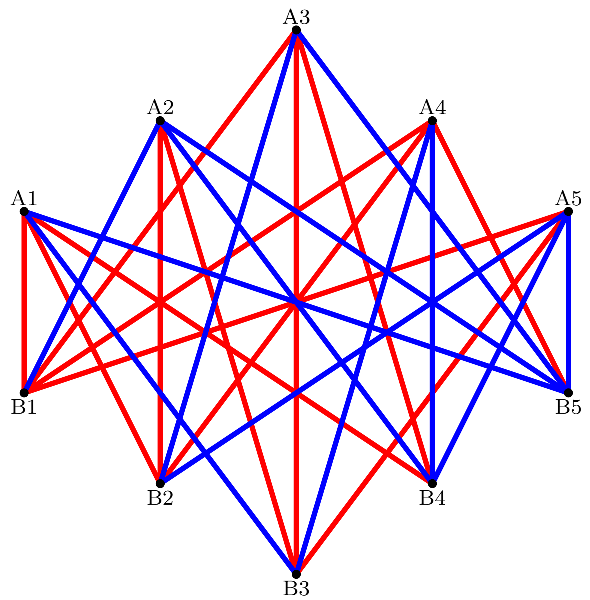

Figure 3.3: $K_{5,5}$ with $K_{2,2}$-red or $K_{2,2}$-blue

Chapter 4

Relationship between Zarankiewicz and bipartite Ramsey numbers

In this chapter, we examine the relationship between Zarankiewicz numbers and bipartite Ramsey numbers, focusing on how the upper bounds of Zarankiewicz numbers can help determine the upper bounds of bipartite Ramsey numbers. The relationship is crucial because it allows us to leverage known results from Zarankiewicz numbers to infer bounds on bipartite Ramsey numbers.

The following proposition from [3] gives connection between the Ramsey numbers and Zarankiewicz numbers. It shows that any method which gives an upper bound for $Z = Z_{a,b}(x,y)$ also gives an upper bound for $R(a,b)$.

Proposition 3. $Z_{a,b}(x,y) < \left\lceil \frac{1}{2}xy \right\rceil$ implies $(x,y) \to (a,b)$, where $\left\lceil \frac{1}{2}xy \right\rceil$ denotes the smallest integer not less than $\frac{1}{2}xy$.

The relationship between these two concepts follows from the related nature of their objectives: while bipartite Ramsey numbers, $R(a,b)$, require coloring the edges of a complete bipartite graph to avoid monochromatic bicliques, Zarankiewicz numbers, $Z_{a,b}(x,y)$, aim to determine the maximum number of edges in a bipartite graph on $x+y$ vertices without $K_{a,b}$ as a subgraph. In general, the evaluation of bipartite Ramsey numbers seems to be challenging, even for the balanced case $R(x,x)$. Furthermore, Goddard, Henning, Oellermann [11] established that upper bound for bipartite Ramsey numbers using upper bound for Zarankiewicz number as

$$Z_{a,a}(x,x) + Z_{b,b}(x,x) < x^2 \implies R(a,b) \leq x \tag{4.1}$$

where $Z_{a,a}(x,x)$ implies the most number of edges in a subgraph of $K_{x,x}$ that has no $K_{a,a}$ subgraph, and $Z_{b,b}(x,x)$ the most number of edges in a subgraph of $K_{x,x}$ that has no $K_{b,b}$ subgraph, so that the total number of edges is greater than the sum of both values. To see how this actually holds, we first consider the red-blue coloring of the edges of $K(x,x)$. Now, if (4.1) holds true for sufficient value of x, it implies that if there can't exist $K_{a,a}$ (red) or $K_{b,b}$ (blue), then,

$$\overbrace{(\text{The number of red edges})}^{\mathbf{R}} \leq Z_{a,a}(x,x) \quad ; \quad \overbrace{(\text{The number of blue edges})}^{\mathbf{B}} \leq Z_{b,b}(x,x)$$

on the other hand, we know that $K_{x,x}$ has x^2 edges then

$$x^2 = R + B \leq \underbrace{Z_{a,a}(x,x) + Z_{b,b}(x,x)}_{< x^2} \implies (\textbf{Contradiction})$$

This implies that every red-blue coloring of the edges of $K_{x,x}$ must contain $K_{a,a}$ (red) or $K_{b,b}$ (blue) resulting $R_{a,b} \leq x$.

Example 4. Consider the known values from [1] for $x = y = 17$, $a = 2$, and $b = 5$, i.e., $R(2,5) = 17$, $\quad Z_{2,2}(17,17) = 74$, $\quad Z_{5,5}(17,17) = 213$. Then,

$$Z_{2,2}(17,17) + Z_{5,5}(17,17) = 74 + 213 = 287 < 17^2 = 289$$

In this example, we actually try to avoid the red ($K_{2,2}$) and blue ($K_{5,5}$) bipartite complete subgraphs. To do that we use there corresponding Zarankiewicz number information i.e., $Z_{2,2}$ edges and $Z_{5,5}$ edges. So, in total we get $(Z_{2,2} + Z_{5,5})$ edges in our bipartite complete graph consisting of red and blue edges. Now, consider $K_{17,17}$, which has 17^2 edges. If we color the edges of $K_{17,17}$ such that there are no monochromatic $K_{2,2}$ in red and no monochromatic $K_{5,5}$ in blue, we see that the total number of edges needed to avoid these monochromatic subgraphs would be $Z_{2,2}(17,17) + Z_{5,5}(17,17) = 287$. Since $287 < 289$, it shows that it is impossible to avoid both a red $K_{2,2}$ and a blue $K_{5,5}$ in $K_{17,17}$, meaning at least one of these subgraphs must exist. Therefore, $R(2,5) \leq 17$.

The following table (4.1), from Collins et al. [1] thesis project, shows the detailed relationship between Zarankiewicz numbers and bipartite Ramsey numbers. In the table, one should focus on the condition $Z_{a,a}(x,x) + Z_{b,b}(x,x) < x^2$. Once this condition is satisfied for a corresponding value of x, we can conclude that $R(a,b) \leq x$. For instance, when $x = 24$, we observe that $Z_{2,2}(24,24) + Z_{6,6}(24,24) = 575 < 24^2 = 576$. This indicates

x	$Z_{2,2}(x,x)$	$Z_{6,6}(x,x)$	$Z_{2,2}(x,x) + Z_{6,6}(x,x)$	x^2
18	81	264	345	324
19	88	293	381	361
20	96	323	419	400
21	105	352	457	441
22	112	383	495	484
23	119	416	535	529
24	127	448	575	576

Table 4.1: Bounds on $Z_{2,2}(x,x)$ and $Z_{6,6}(x,x)$ to establish upper bound for $R(2,6)$

that the condition is met, and thus we get $R(2,6) \leq 24$. By examining the values in the table and checking this condition, we can determine the upper bound for bipartite Ramsey numbers.

Note that, in Table (4.1), the red values are exact values from [2], the blue values are determined by Irving's sharpened bound (see contribution part), and the remaining values are from Collins et al. [1]. Therefore, to determine the upper bound for $R(2,6)$, we can use the upper bounds of $Z_{2,2}(x,x)$ and $Z_{6,6}(x,x)$ for different x values until the condition $(Z_{a,a}(x,x)+Z_{b,b}(x,x) < x^2) \implies R(a,b) \leq x$ is satisfied. Hence, we can conclude that the upper bound of Zarankiewicz numbers helps in generating the upper bound for bipartite Ramsey numbers.

Chapter 5

Contribution

In this project, we have significantly improved the computation of upper bounds for Zarankiewicz numbers by developing an enhanced integer linear programming (ILP) model, utilizing the GLPK solver. Our approach integrates and refines techniques from previous research, including Irving's method (1) and the recursive upper bounds (2) proposed by Dámasdi, Héger, and Szőnyi [2].

5.0.1 Enhanced ILP Model

The primary contribution is the development of a sophisticated ILP model that incorporates advanced concepts and methods. Irving's original approach relies on step-by-step estimates, progressing through smaller parameters to calculate upper bounds. Our model automates this process, streamlining it to reduce manual effort and improve efficiency. Additionally, we have integrated the recursive upper bounds from Damásdi et al., with significant enhancements. In the original method, the presence of a $K_{r,i}$ (a complete bipartite graph with r vertices on one side and i on the other) was required, but not $K_{r,i+1}$. Our model relaxes this requirement, only needing $n[r,i] > 0$, indicating the existence of an r-set with exactly i common neighbors. This weaker assumption broadens the applicability of our method.

5.0.2 Recursive Methods and Constraints

Our model includes a variety of recursive methods as constraints, improving upon Roman's bound (2.3). Both Irving's and Collins et al.'s methods focus on recursive approaches; however, our model extends these ideas by incorporating both maximum and minimum degrees, not limiting the recursion to specific parameters. This flexibility allows us to freely choose any value required for determining the Zarankiewicz number, enhancing the robustness and versatility of our model.

5.0.3 Implementation Details

The implementation begins with importing initial values calculated from Roman's bound. These values serve as the foundation upon which our additional constraints and parameters are applied to maximize the upper bound. The ILP model uses binary indicators to represent the existence of specific degree-based subgraphs, efficiently generating and improving upper bounds within a recursive framework.

The ILP model's variables represent various aspects of the graph structure, including degrees and cardinalities of sets. Constraints ensure proper calculation and relationships between different elements, aligning sums of variables with combinatorial values, balancing set relationships to maintain structural integrity, and eliminating surplus variables to streamline the model. The objective function aims to maximize the variable N, representing the upper bound, thus ensuring the model finds the most efficient solution.

5.0.4 Practical Advantages

Our ILP model generalizes and combines the methodologies from Irving and Damásdi et al. into a single, cohesive framework. This unification not only retains the strengths of each method but also simplifies the overall process. Unlike Irving's step-by-step method, our model provides a more comfortable and less tedious solution. For instance, where Irving's approach calculates lower bounds for $K_{r,i}$ iteratively, our model maintains exact equalities, resulting in more precise and often superior results.

5.0.5 Comparative Analysis

To illustrate the practical application, consider an example initially solved by Irving's method. Using our ILP approach, the same problem is solved more efficiently, demonstrating the model's effectiveness and ease of use. For example, using Irving's lemma (1), we obtained 69 copies of $K_{5,5}$ in $K_{19,19}$ with a Zarankiewicz number of 271 (from [11]). However, our ILP model generated a better result with $Z_{5,5}(19, 19) = 267$. This comparison underscores the advantages of our ILP model, providing clear, automated solutions to different parameter based Zarankiewicz numbers.

5.0.6 Detailed Explanation of the ILP Model

Our ILP model is designed with strong attention to detail, incorporating variables and constraints that capture the essence of graph structure and combinatorial relationships. We assume that there exists a bipartite graph G with partite sets A and B, $|A| = x$, $|B| = y$, which does not contain a $K_{a,b}$ subgraph. The variables and constraints are related to G in a way that we can establish an upper bound on the possible number of edges in G. Here, we detail the components of the ILP model:

5.0.7 Parameters

- `param Z{(x,y,a,b)}`: This parameter defines a four-dimensional matrix used to store specific values (upper bounds for $Z_{s,t}(x,y)$) related to the problem's dimensions. These values are essential for setting up the constraints and objective function.
- `param x, y, a, b`: These parameters represent the cardinalities within the bipartite graph. x and y represent the sizes of the bipartite sets, while a and b indicate the size of the forbidden complete subgraph $K_{a,b}$.

5.0.8 Binomial Coefficients

- `param BINOM{i, j}`: This parameter defines the binomial coefficients, which are crucial for calculating combinations. In the context of the ILP model, these coefficients help in formulating constraints that represent combinatorial relationships in the graph.

5.0.9 Variables

- `var n{r, i}`: This variable represents the number of r-sets in A with exactly i common neighbors (in B). It's a central variable in the model, used to capture the structure of the bipartite graph.
- `var m{r, i}`: Similar to $n_{r,i}$, this variable captures the number of r-sets in B with exactly i common neighbors (in A).
- `var N`: This is the main variable that the objective function aims to maximize. It represents the number of edges in G, that is, the upper bound we are trying to calculate.

- `var p{r, i}, q{r, i}`: These binary variables indicate the presence of specific subgraph structures based on the degree constraints. More specifically, $p\{r, i\} = 1$ if and only if $n\{r, i\} > 0$, and $q\{r, i\} = 1$ if and only if $m\{r, i\} > 0$. They help in formulating the model's logical relationships.

5.0.10 Objective Function

- `maximize MAX_N`: The objective function aims to maximize the sum of $n[1, i] \cdot i$, that is, N. This ensures that the solution we find provides the highest possible upper bound, aligning with our goal of optimizing the upper bound calculation.

5.0.11 Constraints

- `OBJVAL`:

$$N = \sum_{i=0}^{y} n\{1, i\} \cdot i$$

  This constraint ensures that the objective value N aligns with the sum of $n[1, i] \times i$, that is, the sum of the degrees of the vertices in A. It's essential for linking the objective function to the model's variables.

- `DEGREE1 and DEGREE2`:

$$\forall r \in \{1, \dots, a\} \colon \sum_{i=0}^{y} n\{r, i\} = \binom{x}{r}$$

$$\forall r \in \{1, \dots, b\} \colon \sum_{i=0}^{x} m\{r, i\} = \binom{y}{r}$$

  These constraints align the sums of n and m variables with binomial coefficients (that is, the total number of r-sets in A and B), ensuring that the combinatorial relationships are correctly represented in the model.

- `KST`:

$$\forall s \in \{1, \dots, a\}, \forall t \in \{1, \dots, b\} \colon \sum_{i=0}^{y} n\{s, i\} \cdot \binom{i}{t} = \sum_{i=0}^{x} m\{t, i\} \cdot \binom{i}{s}$$

  These constraint balances the relationships between different sets in the bipartite graph. It ensures consistency in how the sets and their degrees interact. Specifically, the number of $K_{s,t}$ subgraphs may be counted in two different ways, seen on the left and the right side of the equality. These constraints are based on Irving's approach, but instead of step by step and independent calculations of lower bounds on the number of $K_{s,t}$ subgraphs for different values of s and t, we use precise equalities simultaneously.

- `ZERO Constraints`: These constraints eliminate surplus variables, streamlining the model and reducing computational complexity.

- `Relational Constraints (P_Rela1, P_Rela2, Q_Rela1, etc.)`:

$$\forall r \in \{1, \ldots, a\}, \forall i \in \{0, \ldots, y\} \colon p(r,i) \leq n(r,i) \leq p(r,i) \cdot \binom{x}{r}$$

$$\forall r \in \{1, \ldots, b\}, \forall i \in \{0, \ldots, x\} \colon q(r,i) \leq m(r,i) \leq q(r,i) \cdot \binom{y}{r}$$

  These constraints ensure that the binary variables p and q accurately reflect the presence of subgraphs. They enforce logical relationships between the variables, ensuring the model behaves as expected: $p\{r,i\}$ can be 1 if and only if $n\{r,i\} > 0$, $q\{r,i\}$ can be 1 if and only if $m\{r,i\} > 0$.

- `KAB`:

$$\sum_{i=0}^{y} n\{a,i\} \cdot \binom{i}{b} = 0$$

  This constraint represents that no $K_{a,b}$ subgraph can be present.

- `UA, UB, U_A, U_B`: These constraints represent the recursive upper bound inspired by Proposition 2 Damásdi et al. If there is an r-set in, say, A such that the vertices in it have exactly i common neighbors in B, then we can apply the same ideas to derive an upper bound on the number of edges:

$$N \leq ri + (y-i)(r-1) + Z_{a-r,b}(x-i,i) + Z_{a,b}(x-r, y-i).$$

  But also, we can give an estimate on the number of edges between M and B (see the proof of the proposition) in one step to derive

$$N \leq ri + (y-i)(r-1) + Z_{a,b}(x-r, y).$$

It is not clear which one is better, so we may take the minimum of the two, denoted by $UA\{r,i\}$ and, symmetrically, by $UB\{r,i\}$. In order to assure that the constraint is applied only if an r-set (in, say, A) with precisely i common neighbors is present (that is, $n\{r,i\} > 0$ or, equivalently, $p\{r,i\} = 1$), we add a huge term to the constraint to make it meaningless when $p\{r,i\} = 0$. (Clearly, xy is the number of all edges in $K_{x,y}$, so the upper bound is valid in all cases.) Hence the constraint is the following:

$$\forall r \in \{1,\ldots,a\}, \forall i \in \{0,\ldots,y\}\colon N \le UA\{r,i\} + (1 - p\{r,i\}) \cdot x \cdot y$$

Similar additional constraints are also introduced, see the script of the IP model.

- `KrimaxA and KrimaxB`: These constraints ensure that if i is maximal such that $p[r,i] = 1$, meaning there exists no $K_{r,i+1}$ in the graph, then we limit N accordingly.

  `KrimaxA`:

  $$\forall r \in \{1,\ldots,a{-}1\}, \forall i \in \{1,\ldots,y{-}1\}\colon N \le Z[x,y,r,i{+}1] + (1 - p[r,i] + \sum_{j=i+1}^{y} p[r,j]) \cdot x \cdot y$$

  This constraint ensures that if there exists no $K_{r,i+1}$ in the graph, then the upper bound N is adjusted accordingly, taking into account the largest i for which $p[r,i] = 1$.

  `KrimaxB`:

  $$\forall r \in \{1,\ldots,b{-}1\}, \forall i \in \{1,\ldots,x{-}1\}\colon N \le Z[y,x,r,i{+}1] + (1 - q[r,i] + \sum_{j=i+1}^{x} q[r,j]) \cdot x \cdot y$$

  Similarly, this constraint ensures that if there exists no $K_{r,i+1}$ in the graph, then the upper bound N is adjusted accordingly, taking into account the largest i for which $q[r,i] = 1$.

To see the complete GLPK mod file script and understand the implementation in detail, please see the appendix. The script is provided there to offer a comprehensive view of the model, including all parameters, variables, constraints, and the objective function as discussed above.

Chapter 6

Conclusion

This thesis project has significantly advanced the computation of upper bounds for Zarankiewicz numbers by developing an enhanced Integer Linear Programming (ILP) model utilizing the GLPK solver. Through the integration and refinement of previous methodologies, including those by Irving [3] and Damásdi et al. [2], the model automates the calculation process, achieving greater accuracy and efficiency.

Fundamentally, this study provides a comprehensive introduction to Zarankiewicz numbers and Bipartite Ramsey numbers, ensuring a thorough understanding of these concepts through clear definitions and visual examples. By revisiting foundational ideas and key bounds, and then generalizing these approaches, the study demonstrates that allowing recursion to move freely within the parameter space can give better results compared to traditional methods. Also, using recursive methods and constraints, our ILP model improves upon existing bounds and provides greater flexibility in parameter selection. This model not only simplifies the calculation of Zarankiewicz numbers but also explain their relationship with Bipartite Ramsey numbers. This connection highlights the practical usage of Zarankiewicz numbers in establishing upper bounds for Bipartite Ramsey numbers, thus providing a unique framework for addressing extremal problems in bipartite graphs.

In many cases, especially for small parameters where Collins et al. [1] did not provide results, our ILP model successfully generates values. Additionally, it sharpens Irving's approach, improves Roman's bound, and generalizes Damásdi et al.'s recursive approach in a more comprehensive manner. The motivation behind this thesis was to generalize the existing bounds, moving beyond specific cases and recursion patterns to create a more flexible and robust model.

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Appendix

For a comprehensive understanding, the complete GLPK input file is included in this section. This file provides insight into the technical implementation of our model. The important steps used in the ILP model are commented to ensure clarity and precision.

Integer Linear Programming Model

The model's implementation in GLPK involves importing initial data, applying constraints, and solving for the upper bound. The recursive constraints are particularly crucial as they enhance the model's capability to handle more complex scenarios efficiently.

A critical component of our implementation is the data file `zar.txt`, which contains pre-calculated values based on Roman's bound for $1 \leq x, y, a, b \leq 20$. This file provides essential initial values required for the recurrence relations used in our ILP model. By leveraging these pre-calculated values, the model can efficiently use the recurrence relations to generate more accurate and robust upper bounds for Zarankiewicz numbers.

```
# data zar.txt;  # Load data from zar.txt
# Reading parameters from zar.txt
# Define parameter Z with appropriate indices
param Z{(x,y,a,b) in 1..20 cross 1..20 cross 1..20 cross 1..20};
# |A|=x, |B|=y

param X;
param Y;
param A;
param B;


param y, integer := Y;
param x, integer := X;
param a, integer := A;
```

```
param b, integer := B;




# Defining the binomial
param BINOM{i in 0..max(x,y), j in 0..max(a,b)} :=
    if (j=0) then 1
    else (if (i >= j) then (round(prod{u in {0..(j-1)}} (i-u)/(j-u)))
          else 0);

# Defining n[r,i] and m[r,i]
var n{r in 1..max(a, b), i in 0..y}, integer >= 0;
var m{r in 1..max(a, b), i in 0..x}, integer >= 0;

# Defining the variable that we want to maximize
var N, integer >= 0;

# Defining binary variables to indicate presence an r-set in A / B with
    degree (exact number of common neighbours) i
var p{r in 1..max(a, b), i in 0..y}, binary;
var q{r in 1..max(a, b), i in 0..x}, binary;

# Objective function: maximize the sum of n[1,i] * i
maximize MAX_N: sum{i in 0..y} n[1,i] * i;

# Constraints

s.t. OBJVAL: N = sum{i in 0..y} n[1,i] * i;

s.t. DEGREE1{r in 1..a}: sum{i in 0..y} n[r,i] = BINOM[x,r];
s.t. DEGREE2{r in 1..b}: sum{i in 0..x} m[r,i] = BINOM[y,r];
s.t. KST{s in 1..a, t in 1..b}:
    sum{i in 0..y} (n[s,i] * BINOM[i, t]) = sum{i in 0..x} (m[t,i] *
        BINOM[i, s]);
s.t. KAB: sum{i in 0..y} n[a,i] * BINOM[i,b] = 0;

# Zero constraints for the surplus variables
s.t. ZERO1{r in (a+1)..max(a, b), i in 0..y}: n[r,i] = 0;
s.t. ZERO2{r in (b+1)..max(a, b), i in 0..x}: m[r,i] = 0;
s.t. ZERO3{r in (a+1)..max(a, b), i in 0..y}: p[r,i] = 0;
```

```
s.t. ZERO4{r in (b+1)..max(a, b), i in 0..x}: q[r,i] = 0;


# We want p[r,i] = 1 if and only if n[r,i] > 0, and analogously for q
    and m.
# Thus the value of n[r,i] should not exceed p[r,i] times BINOM[x,r],
    but should be at least p[r,i], and analogously.
s.t. P_Rela1{r in 1..(a-1), i in 0..y}: n[r,i] <= p[r,i] * BINOM[x,r];
s.t. P_Rela2{r in 1..(a-1), i in 0..y}: n[r,i] >= p[r,i];
s.t. Q_Rela1{r in 1..(b-1), i in 0..x}: m[r,i] <= q[r,i] * BINOM[y,r];
s.t. Q_Rela2{r in 1..(b-1), i in 0..x}: m[r,i] >= q[r,i];



# We add the upper bound constraint for recursive upper bound
    calculation
# UA is valid if there is an r-set in A with degree exactly i, that is,
    p[r,i]=1; UB is valid iff q[r,i]=1.
param UA{r in 1..(a-1), i in 1..y} :=
    if (i<y) then min(r*i + (y-i)*(r-1) + Z[x-r,i,a-r,b] + Z[x-r,y-i,a,b
        ], r*i + (y-i)*(r-1) + Z[x-r,y,a,b]) else r*y + Z[x-r,y,a-r,b];

param UB{r in 1..(b-1), i in 1..x} :=
    if  (i<x) then min(r*i + (x-i)*(r-1) + Z[y-r,i,b-r,a] + Z[y-r,x-i,b,
        a], r*i + (x-i)*(r-1) + Z[y-r,x,b,a]) else r*x + Z[y-r,x,b-r,a];

# Constraints for i<x and i< y; meaningful only of p[r,i] = 1 (or q[r,i]
     = 1)
s.t. U_A{r in 1..(a-1), i in 1..(y-1)}:
    N <= UA[r,i] + (1 - p[r,i]) * x * y;

s.t. U_B{r in 1..(b-1), i in 1..(x-1)}:
    N <= UB[r,i] + (1 - q[r,i]) * x * y;

# Constraints for the case when i = y
s.t. U_A_y{r in 1..(a-1)}:
    N <= UA[r,y] + (1 - p[r,y]) * x * y;

# Constraints for the case when i = x
s.t. U_B_x{r in 1..(b-1)}:
    N <= UB[r,x] + (1 - q[r,x]) * x * y;
```

```

# Constraints if i is maximal such that p[r,i] = 1; that is, there
    exists no K_{r,i+1} in the graph
s.t. KrimaxA{r in 1..(a-1), i in 1..y-1}:
  N <= Z[x,y,r,i+1] + (1 - p[r,i] + (sum{j in i+1..y} p[r,j]))*x*y;

s.t. KrimaxB{r in 1..(b-1), i in 1..x-1}:
  N <= Z[y,x,r,i+1] + (1 - q[r,i] + (sum{j in i+1..x} q[r,j]))*x*y;


end;
```

Code 6.1: Integer Linear Programming Model